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HUMAN RESOURCE MANAGEMENT SYSTEM

A Capstone Project
Presented to the Faculty of
College of Computer Studies
Carlos Hilado Memorial State University,
Bacolod City

In Partial Fulfillment
of the Requirement for the Degree
Bachelor of Science in Information Systems

by:

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December 2025



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ABSTRACT

This project aims to develop a modern human resource management system designed to meet the needs of the Local Government. The system is designed to improve the efficiency of Human Resource management in Bacolod City by streamlining traditional manual processes. It can be accessed on both web and mobile platforms, allowing employees to easily manage HR-related information, including leave, benefits, and retirement. The development of this system followed the Software Development Life Cycle (SDLC), which includes the phases of planning, analysis, and design. Implementation, testing, and deployment to ensure that it meets the end-users' requirements. System evaluation was done using the post-study system usability questionnaire, which then measured the dimensions of system usefulness, information quality, and interface design. The results indicated a very satisfying relationship, as respondents agreed that the system is both intuitive and practical, thereby adding productivity to the workplace. Accordingly, the HRMS will contribute to reducing paper-driven processes in human resource operations at the local government level, resulting in improved service delivery and increased overall efficiency.

Keywords: Human Resource Management System, Bacolod City LGU, system development, process automation, SDLC, PSSUQ, user satisfaction.

CHAPTER I

INTRODUCTION

Background of the Study

The Bacolod City Human Resource Management Services cannot work efficiently because it still depends on manual, paper-heavy methods. This reliance often leads to lengthy delays in approval. It makes it very difficult for staff to access vital HR assistance, such as applying for leave, tracking their benefits, or filing for retirement. As the public continues to demand faster and more reliable government services, updating our existing systems has become a critical priority.

This transition to digital solutions is strongly justified by the proven success seen in other cities. For example, Quezon City demonstrated this when its new HR system made applying for leave significantly faster and easier (Gonzales, 2021). Similarly, Zamboanga City now utilizes a digital HR system that facilitates a smoother onboarding process for new employees and enables managers to make more informed decisions (Delos Santos-Suñga & Moreno, 2024).

Even here in Bacolod, a 2022 survey by the University of St. La Salle confirmed that most local government employees would prefer a web-based HR system because it is easier to access and much more convenient.

The new system is primarily designed to automate the basic HR tasks that are currently performed manually and on paper, and move them online. This change will

significantly reduce the paperwork and trouble for our employees, resulting in improved service for everyone. We will build the system carefully, using the trusted, step-by-step approach for software development (called the SDLC). When it is finished, we will assess its helpfulness and ease of use by asking the staff to complete a feedback survey called the PSSUQ.

Statement of the Objective

General Objective

The objective of this study is to create a web-based Human Resource Management System (HRMS). This system is intended for use by the Office of Human Resource Management Services of the City Government of Bacolod to improve efficiency in managing leave requests, retirement procedures, and employee benefits. The system is designed to offer an accurate and efficient method for monitoring remaining leave balances, terminal leave benefits, and other employee entitlements. This should facilitate rapid access to information and expedite decision-making for Human Resource staff.

Specific Objectives

This study specifically aims to:

1. To develop an automate processing of employee leaves, benefits, and retirement within the Bacolod City LGU to improve HR operations.
2. A real-time monitoring system that tracks leave status, leave balance entitlements for all active employees.

3. Develop a reporting dashboard capable of generating accurate and downloadable summaries of leave applications and leave balances, as well as retirement lists.
4. Minimize physical employee transactions and administrative walk-ins to the HRMS office.
5. This will be automatic; the input for leave type, leave details, and inclusive accuracy will be provided through a structured digital form.
6. To generate a printable form and leave an application form based on the employee's information.
7. Evaluate system acceptability by analyzing PSSUQ results collected within one month after implementation.

Framework of the Study

Figure 1 shows the conceptual framework used in the study. The conceptual framework will serve as a guide in the system's development process.

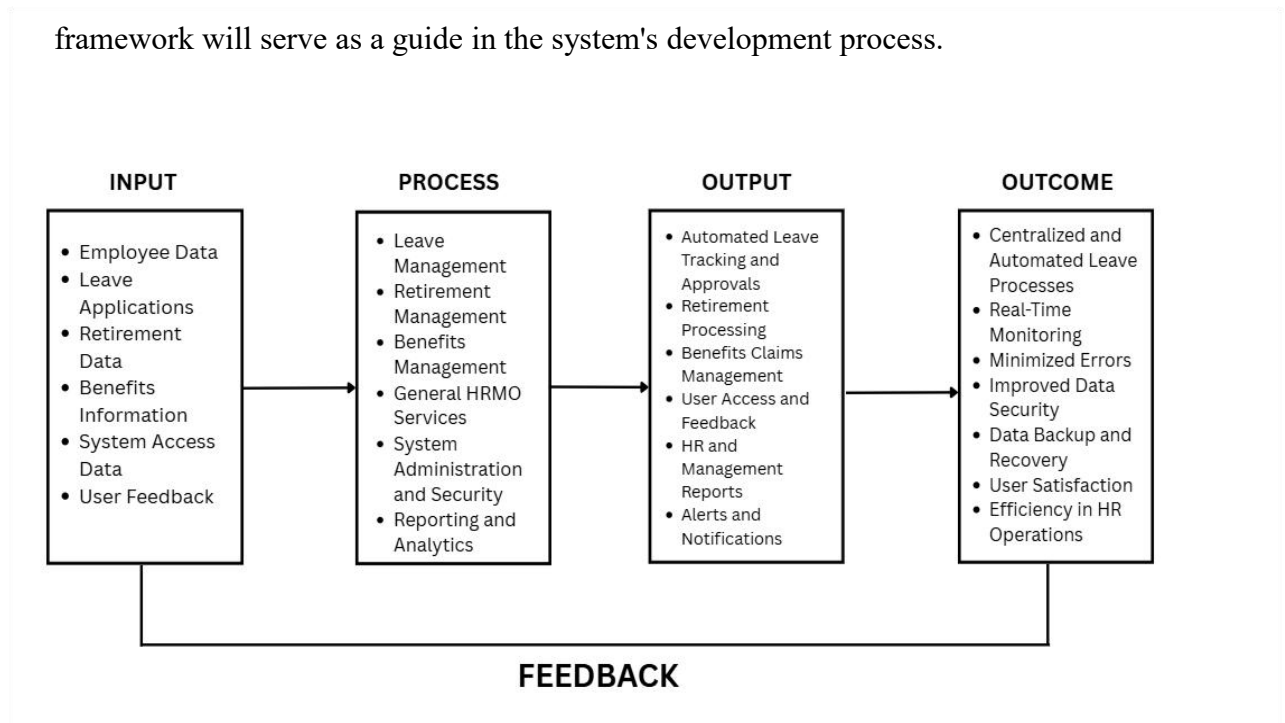


Figure 1. Conceptual Framework of the Study

Scope and Delimitation of the Study

Scope

The study focus is on developing a Human Resource Management System (HRMS) designed to automate leave filing, benefit tracking, and retirement preparation. The scope includes an online portal where employees can file leave applications remotely. To comply with documentation standards, the system features an auto-fill function that populates the required data into standard templates—including official **GSIS Retirement Forms**—which the employee can then print for physical signature. The system also tracks leave credit balances, monitors eligibility for **Terminal Leave** and **Loyalty Awards**, and provides a centralized leave calendar and notification system to keep employees updated on their application status.

Delimitation

The system is subject to specific administrative boundaries to maintain the separation of duties. The system will not integrate the employee's attendance and payroll. The system will not perform any salary computations, payroll processing, or financial transactions; it will also not cover other HR operations, such as recruitment, training, or employee discipline records. The HRMS does not have a direct data link or integration with the government service insurance system(GSIS). Its role is strictly to prepare the necessary documents and guides; the actual submission of rental claims and the final approval of funds must be processed independently by the employee through the GSIS office.

Significance of the Study

Results of the study may be significant on the following:

The regular employees, this system is highly significant as it provides the convenience of remote filing. By applying, the CSC Form 6 application will be automatically filed. The system will reduce manual handwriting errors for the employee. Furthermore, it serves as a reliable portal where they can download blank GSI retirement forms, ensuring they have immediate access to the correct templates needed for their retirement preparation.

Hr staff, the system will serve as a gatekeeper for the leave benefits and retirements. It is significant to the hr staff as it automates the tracking of leave credits and notifies employees that they are eligible for Loyalty Awards and Terminal Leave.

The hr officer, the study provides the hr officer with a verified date for certifying the official record. Since the HRMS tracks bonuses in real time, the HR Officer can sign the printed form as the Authorized Officer with the assurance that the leave credits listed are accurate and up-to-date.

The department head represents the authorized official approver for leave benefits, receiving a clean, auto-filled document that has already been vetted for balance efficiency by the system and HR staff, allowing for a more informed and faster approval process.

To the Researchers: This study provides a practical framework for researchers designing "hybrid" systems. It illustrates how to combine digital

automation (auto-filling internal leave forms) with resource management (downloadable external government forms) while maintaining compliance with traditional "wet signature" protocols.

Operational Definition of Terms

Authorized Officer: The designated HR personnel responsible for validating the accuracy of the employee's calculated leave balance and providing official certification on the printed CSC Form 6.

Authorized Official: This designation refers to the Department Head or equivalent authority who has the ultimate decision-making power in granting or denying leave on all applications submitted through the digital system.

Auto-fill Function: This important feature of the HRMS automatically retrieves and populates the employee's relevant personal and leave entitlement data directly onto the digital CSC Form 6 template; hence, the document is instantly prepared for printing and subsequent submission.

CSC Form 6: The official documentary requirement of the CSC, also titled "Application for Leave". In the HRMS, this document is the primary output that the system generates and populates.

Document Repository: A repository module that is available to the employees within HRMS, used currently to provide easily downloadable, blank, official retirement application forms from GSIS to the user for their manual preparation.

Government Service Insurance System (GSIS): The external government agency tasked with managing social insurance for employees in the government. The HRMS

provides employees with access to the necessary blank retirement forms through the Document Repository.

Leave Balance Efficiency refers to the critical step in the process where HR actually confirms the available accruals of leave for an employee against the duration requested in their application before giving final approval. This is important to ensure leave policies are followed, and unauthorized deficits are not accrued.

Loyalty Award An internal HRMS benefit is automatically tallied and provided to employees upon attaining certain organizational milestones, based on their years of continuous service.

Regular Employee A primary end-user who uses the system to initiate formal leave requests, view current and projected earned leave, and retrieve necessary retirement or separation forms.

Terminal Leave Total accumulated and unused leave credits of an employee in their monetary equivalent, which is computed and paid to the employee upon official separation or retirement from the service.

Wet Signature The physical, ink-based signing of printed, auto-filled system forms by authorized personnel, including the HR Officer and Department Head, is strictly required for official compliance with government audits and regulations.

CHAPTER II

REVIEW OF RELATED LITERATURE

Application of Time – Based One Time Password (TOTP) Algorithm For Human Resource E- Leave Tracking Web App

This study examines the application of the Time-Based One-Time Password (TOTP) algorithm to enhance the security of a Human Resource Electronic Leave Tracking Web Application. TOTP is a type of two-factor authentication (2FA) that adds an extra layer of security when logging in, helping to protect user accounts. With TOTP, employees can log in using a one-time password that changes every few seconds. This helps stop unauthorized access and keeps leave requests safe. By using this method, the system becomes more reliable and secure for HR departments (Plata & Calpito, 2020).

Digital Human Resource Management: Prospects & Challenges for Garments Industries in Bangladesh

The abstract highlights the growing adoption of Digital Human Resource Management (D-HRM) in countries such as Bangladesh. It explains how automating HR tasks helps companies save time and money, improve hiring through social media, and enhance their image using HR Information Systems (HRIS).

The study examines the use of digital HRM in Bangladesh's garment industry, which is one of the world's largest exporters of ready-made clothing. A group discussion was held with eight top garment factories to learn about their digital HRM practices. As countries like China move forward with automation, Bangladesh needs to enhance its digital HR systems to remain competitive. The study demonstrates that digital HRM can effectively manage workers and support the country's position in the global market.

Employee Leave Management System

The Employee Leave Management System streamlines leave requests, monitoring, and approvals, eliminating tedious, time-consuming, error-prone, and paperwork-intensive manual processes. Institutions typically handle leave through paper requests that pass through several approval levels, resulting in inefficiency in the process. To counter this, an automated system was created using a three-tier software framework based on CSS, JavaScript, HTML, MySQL, and PHP, hosted on a Windows OS. The system enhances efficiency by minimizing errors, paperwork, and processing time. (Adamu, 2020).

Leave management information system using insides DPS software for efficiency of human resources management

This research emphasizes the significance of leave management in achieving higher organizational performance and employees' wellbeing. Effective utilization of leaves enhances productivity, family cohesion, and a high-performance work culture. Ineffective leave management can lead to burnout and increased employee turnover. To address these shortcomings, Management Information Systems (MIS) are essential for planning, implementing, monitoring, and evaluating leave procedures. An efficiently designed Leave MIS allows efficient leave requests, approvals, and compliance with company policies, which finally leads to employee satisfaction. In this study, the author suggests adopting the InsideDPS software, which has been successfully implemented in PT. Surabaya Docs and Shipping, to enhance the efficiency of leave management. The system enables employees to apply for leave with ease, fostering improved

communication between employees and management while maintaining uninterrupted business operations. (Wahjono et al., 2020).

A Study on Post retirement pension scheme evaluation for government employee in India

Indian government employees are entitled to terminal benefits and a retirement pension upon reaching the prescribed superannuation age. Government recruitment is conducted through competitive exams that involve stringent verification mechanisms, and government employees serve the public interest with fixed remuneration and allowances. They have key roles to perform in functioning within the government, as government employees continued working during the COVID-19 pandemic despite lockdowns. Although government employees make up a small percentage of India's population, they contribute a significant percentage of taxes as direct taxpayers. The Old Pension Scheme (OPS) for government employees appointed before 2004 ensures post-retirement income protection and benefits for surviving family members. The Indian government established the "OASIS" project in 1999 to assess policies on income security for old age, leading to the introduction of the National Pension Scheme (NPS) in 2004. The Pension Fund Regulatory & Development Authority (PFRDA) was established in 2003 to oversee this change, shifting new government appointees to a contributory pension plan. (Mahalakshmi & College, 2023)

Human resource technology disruptions and their implications for human resource management in healthcare organizations

Concern among policymakers and the public regarding existing and impending large-scale staff shortages is increasing. Enhancing Human Resource (HR) practices, as

well as implementing HR technologies such as Human Resource Information Systems (HRIS), which are capable of gathering, storing, and reporting manpower data, are commonly viewed as the solution to this issue. In fact, illustrations drawn from other sectors suggest that HRIS can assist in initiating or controlling, as well as provide continuous knowledge regarding the entire career life of a worker. Nevertheless, few previous studies that have written about technology or its ramifications for the future of work have addressed health organizations, and those that have received insufficient attention within the health literature. In addition, the few contributions that have been made have either centered on a specific category of technology or primarily highlighted the impact of automation on the work of health professionals.

In conclusion, health organizations and the health and HR practitioners working within them must responsibly utilize HRIS, striking a balance between the motivation for innovation, productivity, and efficiency, and respect for all possible legal, ethical, and compliance considerations. Additionally, they must consider the significance of HRH's wellbeing and satisfaction. Keywords: eHealth, Human resources, Health care management, Information systems, Human resource information systems, HRIS, People analytics (Tursunbayeva, 2019)

Still our most important asset: A systematic review on human resource management in the midst of the fourth industrial revolutions

Digital transformation is changing society as a whole and industry specifically to such an extent that Industry 4.0 has become widespread and consolidated by now. Nevertheless, there is a danger that an overly accentuated consideration of technology can cause us to forget the importance of human resources. Among others, the European

Commission has called for increased attention by research and industry to the human factor. A greater need exists to theorize regarding the human role in a digitalized industry. Establishing a direction for improving the human factor in a digitized industry begins with understanding what is recognized in the literature regarding human resource management in Industry 4.0. The findings provide researchers and managers with an up-to-date and comprehensive view of studies in human resource management within the industry 4.0 context, along with future research directions. (Ammirato et al., 2023)

Early retirement incentive programs as a human resource restructuring in public sector

The concept of workforce downsizing can be defined as decreasing the number of employees, or more specifically, as part of overall attempts to restructure human capital in public agencies. Early Retirement Incentive Programs (ERIP) are an achievable option for organizations aiming to downsize their staff. For the ERIP program to succeed, its coordinator must be knowledgeable about the organizational objectives and aims that the agency is seeking to achieve. Originality/value – Human resources policies regarding downsizing public administration employees should be more relevant to both those who remain outside the organization and those who remain at work, thereby reducing the adverse psychological, administrative, and economic impacts. This can be achieved through a policy known as early retirement incentive programs.

Keywords Human Resources, Retirement, Downsizing, Restructuring, Older worker, Early Retirement Incentive Programs (ERIP) (Shabat, 2020)

Effects of early retirement policy changes on working until retirement: Natural Experiment

Several European countries have implemented pension reforms to raise the statutory retirement age, aiming to increase the labor supply. However, not every older worker can or wants to work until an advanced age. Drawing on a country-wide register record of labor market transitions, we investigated in this natural experiment the effect of an unanticipated change in the Dutch pension system on labor market activities among older workers. Specifically, we investigated labor market transitions over a 5-year period in two nationwide cohorts of Dutch 60-year-old workers, following them until they reached the age of retirement ($n = 23,703$). We compared transitions for the cohort that was still eligible to claim early retirement benefits with a cohort that was no longer eligible to do so. Results showed the pension reform effectively pushed work participation to the statutory retirement age (82% vs. 61% at age 64), as well as to a higher percentage of unemployment benefits in the 1950 birth cohort (2.0–4.2%) compared with the 1949 birth cohort (1.4–3.2%). Thus, while ambitious pension changes can be beneficial for the labor supply, the adverse effects should be considered, especially since other studies have confirmed a link between unemployment and poor health.(Boot et al., 2019)

The relationship between workplace psychosocial environment and retirement intentions and actual retirement: a systematic review

Psychosocial work factors are probable predictors of retirement intentions and actual retirement. A systematic review examined the effects of psychosocial work factors

on retirement intentions and actual retirement in the general population. This did not extend to individuals who were sick and knew they were sick or were in receipt of disability pension. Relevant articles were identified through a search of PubMed, PsycINFO, and Web of Science databases up to December 2016. Longitudinal and cross-sectional articles that measured work psychosocial characteristics in relation to plans for retirement or actual retirement were included. Articles were screened by title and abstract before data extraction was carried out on full texts from a previously defined extraction sheet. Forty-six articles had relevant evidence. Elevated job satisfaction and elevated job control were linked with subsequent retirement plans and retirement. There was no consistent evidence to support an association between job demands and subsequent retirement plans or retirement. We conclude that policies to extend working lives should enhance the job control available to older workers. (Browne et al., 2019)

Analysis of decision support system on situational leadership styles on work motivation and employee performance

This research will examine the outcome of employee performance appraisal using a decision support system (DSS) on situational leadership with work motivation as a mediator of employee performance. The utilization of this type of quantitative research, with a saturated sample technique, will make the research more relevant and objective in presenting the results. Data analysis using smart PLS 3.0 led to the conclusion that the decision support system (DSS) of situational leadership had not been effective in enhancing employee performance. It can be concluded that technology's role in semi-structured decision-making needs to be grounded in adequate data to enhance employee motivation and performance effectively. Indeed, leadership continues to be applied in

making unstructured choices without considering the data in the performance system. Performance measures are often inadequately accorded attention, resulting in poor outcomes from this research. (Wuryani et al., 2021)

Optimization of Human Resource File Information Decision Support System based on cloud computing

With the rapid development of the science and technology era, human resources and knowledge resources have become essential elements in the growth and development of enterprises. Under such circumstances, it is extremely necessary to build a human resources data pool and conduct data mining on it, so that high-quality and high-quantity information can be extracted to offer assistance for managers' decision-making. In this paper, the Human Resource Archive Information Decision Support System (DSS) is designed for various management and decision-making tasks, leveraging the advantages of cloud computing, including scalability, high reliability, flexibility, and extensibility. According to the analysis of "cloud computing" benefits in resources integration and sharing etc., based on this system is developed by employing the foundation of the data acquisition layer, network service support layer, cloud computing support layer, data standardization conversion layer, system application layer, system layer, Decision Support layer and so on seven layer architecture, its characteristics and functions of each layer structure, the working mode and working mode of the Decision Support System (DSS) are elaborated in detail. The system compensates for the shortcomings of conventional archive management, including the scarcity of data resources, the inability to achieve isomorphism, and the uniform processing of data from various data sources. (Cai & Chen, 2021)

Related Systems

Table 1. Comparison of the Related Systems

Features	Human Resource Management System	Employee Leave Management System	Digital Human Resource Management: Prospects & Challenges for Garments Industries in Bangladesh	Leave management information system using insides DPS software for efficiency of human resources management
Online Leave Application	✓	✓	✗	✓
Leave Approval Workflow	✓	✓	✗	✓
Leave Balance Tracking	✓	✓	✗	✗
Leave Calendar	✓	✓	✗	✗
Retirement Process	✓	✗	✗	✗
Retirement forms	✓	✓	✗	✗
Retiree Database	✓	✗	✗	✗
Online Benefits eligibility	✓	✗	✗	✗
Benefits information Portal	✓	✗	✗	✗
Dependents management	✓	✗	✗	✗
Employee Information Portal	✓	✓	✓	✗

Human Resource Management System

Online Forms and Documents	✓	✓	✓	✓
Announcements and Notifications	✓	✓	✓	✗
Feedback and Support	✓	✓	✗	✓

Synthesis

The proposed development of a web-based Human Resource Management System (HRMS) for the Bacolod City Government is a necessary response to the significant barriers created by the continued reliance on manual, paper-based processes in their Human Resource Management Services (HRMS). These outdated systems currently lead to processing delays, data inaccuracies, and limited access to critical HR functions, such as leave applications and retirement processing, despite Bacolod City's emergence as a leading local government unit (LGU) that is embracing digitalization. This capstone project aims to automate core HR functions, including employee leave processing, benefit tracking, and retirement preparation, providing real-time monitoring of leave status and balances while minimizing administrative walk-ins. The system is designed as a hybrid solution, automatically generating and populating the official CSC Form 6 while providing access to GSIS Retirement Forms, thereby enhancing efficiency for both employees and HR staff. However, to comply with government audit standards, the system is deliberately delimited, as it will require physical signatures (wet signatures) on printed forms and will not perform the actual monetary computations for terminal

leave or retirement benefits, which remain the responsibility of the Accounting Department and the GSIS, respectively.

CHAPTER III

METHODOLOGY

This chapter outlines the research design and development process for the City Government of Bacolod's Human Resource Management System (HRMS). The researchers adopted the **Agile Model**, a flexible and iterative software development approach that promotes continuous collaboration between developers and users to deliver a system that accurately meets user needs.

System Development Model

The Agile Model emphasizes iterative progress through small, manageable increments, allowing for regular user feedback and adaptation throughout the development cycle. This method is ideal for projects where requirements may evolve or require refinement as development progresses. Collaboration with stakeholders—HR personnel, employees, and researchers—was a key factor in guiding the development.

The Agile process in this study consisted of the following stages:



Figure 2. Agile Model

Plan

The research team convened an initial meeting to deliberate on selecting barangays 22 and city government for their study. After careful consideration, the group chose City Government Human Resource Management. Before conducting the interviews, the researchers drafted a letter outlining the purpose of the visit and the intended discussions with the City Government HR Officer Jose MA. Daniel Evidente.

The First interview took place in City Government HRMS. The researcher developed a potential system title for HRMS.

These proposed system titles and a GAP analysis were subsequently presented for review. After a thorough evaluation, the Human Resource Management System (HRMS) for the Bacolod City Government Center was selected as the most suitable proposal. This decision marked a significant milestone in the research process.

Design

The system developers sketched and outlined the data flow diagram of both levels and entity relationship diagram on paper before creating it on Lucidchart, a web-based diagramming application. The data flow diagram is for the system flow, and the entity relationship diagram is for creating and analyzing the relationship between entities and the databases. However, the architectural diagram was directly created on Lucidchart.

The architectural diagram is for mapping out the physical execution of the system. This phase provides a basis for the developed system to be used and reviewed by the HRMS.

Development

The Development phase focuses on coding and building the initial functional prototype of the HRMS, often targeting a specific set of features (a sprint goal). Key functionality, such as the online leave application system, leave balance tracking, and the Auto-fill Function for forms, is engineered and integrated. The goal is to rapidly produce a working increment of the system that is ready for user evaluation and testing.

Testing

After the system has been developed, it is placed and arranged in the city government HRMS to allow users to test and interact with it. It involves an evaluation of the system's functionality to deliver a practical user experience. Before the deployment phase, testing must be done to ensure the system's quality.

Deployment

The main goal of deploying is to determine whether the system is approved. Otherwise, the developers will continue to improve the system and go back to the city government HRMS from time to time until the expectations in the system created are met accurately. This phase requires thorough and detailed planning to implement the system.

Review

The Review phase, conducted shortly after deployment, is dedicated to formally assessing the system's successful implementation and measuring user acceptance. This stage fulfills the critical objective of evaluating overall system acceptability in the live environment. The team gathers direct feedback on the delivered functionality to confirm

it meets the needs of users (Employees, HR Staff, etc.) and to identify any remaining issues or areas for improvement that will be prioritized in the next planning cycle.

Operational Feasibility

Front End

HTML

HTML is used to create the entire article and the system's structure.

This language is used in the system's front end. When a user accesses the system, everything he sees is coded in HTML, and the programmer uses PHP to construct the website's back end.

CSS

Its user interface is based on Cascading Style Sheets (CSS), so everything remains clean and organized. By using CSS, the layout incorporates standard layouts, color schemes, and typography, thereby making the system more readable and usable. No matter what device you are using—phone, tablet, or computer—the system adjusts smoothly to give you a simple and easy experience. This thoughtful design ensures you can focus on your work without being distracted by messy or confusing layouts.

Javascript

JavaScript is needed to give an interactive and dynamic user interface. Real-time updates, client-side form validation, interactive elements such as tabs and modals, and smooth navigation are all enabled through it. JavaScript is incorporated into PHP and CSS to give a smooth and interactive user experience.

Back End

PHP

The logic and functionality of HRMS are crafted in PHP (Hypertext Preprocessor), which is a core server-side programming language. It provides the ability to automate leave request processing, retirement processes, and benefit calculation. PHP enables the modification of the system to meet the specific needs of Bacolod City LGU, while ensuring compliance with local HR laws and practices.

MYSQL

All employee details, leave applications, retirement business, and benefits are predominantly held in MySQL, the application database engine. MySQL is an RDBMS providing easy scalability for growing

corporate data, fast query execution, and data integrity. It also provides secure storage and multi-user access.

jQuery

The jQuery library is based on concise and easier-to-understand code. It can reduce the time required to implement the system by utilizing simple syntax and open coding standards. jQuery excels at dealing with cross-browser concerns. jQuery has helped resolve these challenges. This eliminates the need to write separate programs for different browsers. It leads to increased efficiency and fewer headaches.

Technical Feasibility

Hardware Specification

Table 2. Minimum Hardware Requirements

Hardware	Requirement Specification
Processor	AMD Ryzen 3 7320U with Radeon Graphics 2.40GHz and above processor
Memory	8 Gigabytes
Storage	475 Gigabytes and above
Graphics Card	Video Card with a display resolution of at least 1024x768
Network	15/50 Mbps
Printer	Dot Matrix/Inkjet/Laser

Software Specification

Table 3. Minimum Software Requirements

Software	Requirement Specification
Operating System	Windows 11
Browser	Google Chrome, Microsoft Edge, Mozilla Firefox

Architectural Diagram

Figure 3 depicts the Architectural Diagram of the Human Resource Management System (HRMS) built for the Bacolod City LGU, which is divided into modular components to provide a scalable, efficient, and secure digital environment for managing essential HR functions.

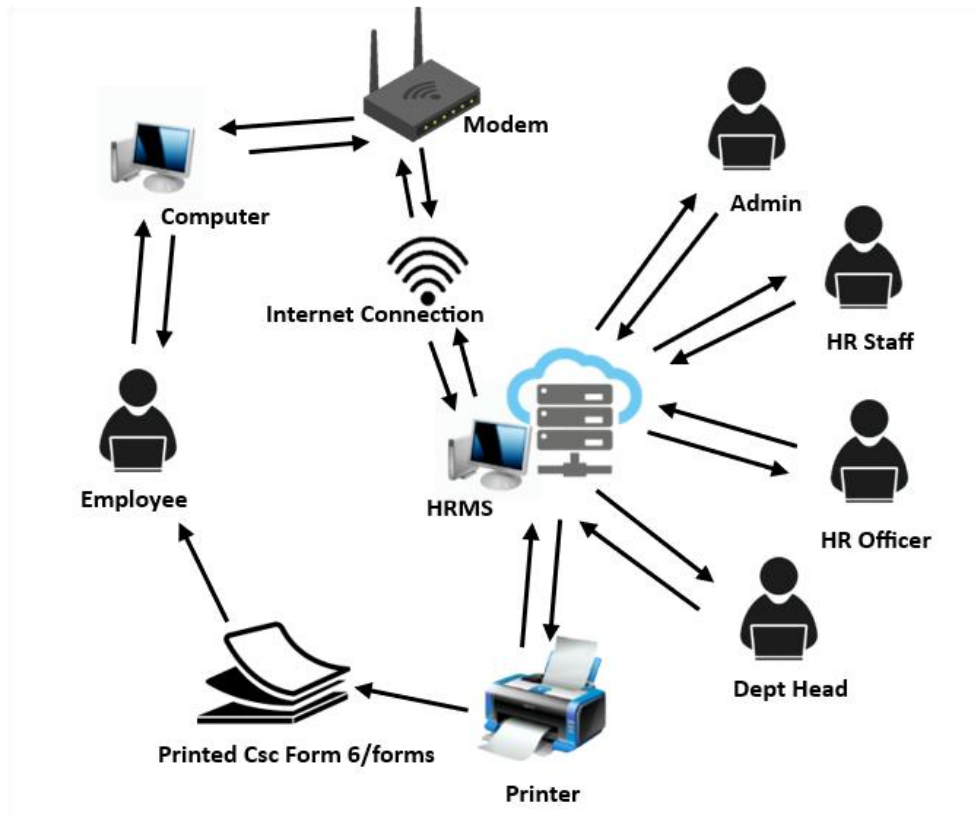


Figure 3. Architectural Diagram

Feasibility Schedule

Work Breakdown

The Work Breakdown Structure (WBS) for the Human Resource Management System (HRMS) project provides a detailed and hierarchical decomposition of all essential tasks, activities, and deliverables required to complete the system. It visually represents the entire structure of the project—starting from the primary objective at the top level and breaking it down into progressively smaller, more manageable components.

ACTIVITY	Assigned To	PLAN START	PLAN DURATION	START DATE	END DATE	ACTUAL START	ACTUAL DURATION	PERCENT COMPLETE
PLAN	(Follow the sequence to rollout)	#1	#2			#3	#4	#5
Brainstorming	ALL Members	1	6	10/2/2025	15/2/2025	3	5	97%
Project Selection	ALL Members	21	10	2/3/2025	11/3/2025	21	12	100%
Document Analysis	ALL Members	23	5	4/3/2025	8/3/2025	23	5	100%
Focus Group	ALL Members	6	14	15/2/2025	28/2/2025	6	14	100%
Interview	Project Leader	21	20	2/3/2025	21/3/2025	23	16	95%
Observation	Project Leader	21	20	2/3/2025	21/3/2025	23	16	95%
Survey/Questionnaire	ALL Members	121	94	10/6/2025	11/9/2025	237	3	100%
DESIGN								
Story Mapping	ALL Members	41	3	22/3/2025	24/3/2025	40	5	100%
Use Case Diagram	Project Leader and System Analyst Programmer	51	5	1/4/2025	5/4/2025	51	255	100%
Wireframes and Mockups	ALL Members	44	7	25/3/2025	31/3/2025	44	253	100%
ER Diagram	System Analyst/Programmer	56	3	6/4/2025	8/4/2025	56	256	100%
Data Flow Diagram	Project Leader and System Analyst Programmer	59	3	9/4/2025	11/4/2025	59	258	100%
DEVELOPMENT								
Coding	System Analyst/Programmer	31	92	12/3/2025	11/6/2025	92	233	100%

Figure 4. Gantt Chart (Part 1)

ACTIVITY	Assigned To	PLAN START	PLAN DURATION	START DATE	END DATE	ACTUAL START	ACTUAL DURATION	PERCENT COMPLETE
TESTING	(Follow the sequence to rollout)	#1	#2			#3	#4	#5
Requirements Verification	ALL Members	123	5	2/9/2025	6/9/2025	123	2	75%
Test Plan	ALL Members	140	10	19/9/2025	28/9/2025	139	8	100%
Test Environment Setup	Project Manager	140	15	19/9/2025	3/10/2025	142	10	100%
Test Execution	ALL Members	154	7	3/10/2025	9/10/2025	155	4	100%
Requirements Reverification	ALL Members	161	3	10/10/2025	12/10/2025	162	6	80%
Retest Environment Setup	Project Manager	163	4	12/10/2025	15/10/2025	164	1	100%
Retest Execution	ALL Members	166	5	15/10/2025	19/10/2025	168	2	100%
DEPLOYMENT								
The Draft/Initial Plan	ALL Members	171	3	20/10/2025	22/10/2025	171	1	75%
Business Communication	Project Manager	173	4	22/10/2025	25/10/2025	174	3	100%
Deployment Plan	ALL Members	176	5	25/10/2025	29/10/2025	176	4	100%
Rollback Plan	ALL Members	181	5	30/10/2025	3/11/2025	183	6	100%
Deployment Exercise	ALL Members	186	7	4/11/2025	10/11/2025	188	3	80%
Going Through Tasks	ALL Members	193	7	11/11/2025	17/11/2025	193	1	100%
Releasing	ALL Members	200	3	18/11/2025	20/11/2025	200	10	100%
REVIEW								
Major Modifications	Programmer	202	10	20/11/2025	29/11/2025	202	7	90%
Routine Changes	Programmer	203	10	21/11/2025	30/11/2025	203	4	100%
Emergency Changes	Programmer	204	10	22/11/2025	1/12/2025	204	8	90%
Software Patches	Programmer	205	10	23/11/2025	2/12/2025	205	9	90%

Figure 5. Gantt Chart (Part 2)

Cost Benefit Analysis

The primary goal of the Cost Benefit Analysis (CBA), a crucial decision-making tool for assessing the viability and financial rationale of the Human Resource Management System (HRMS) project for Bacolod City LGU, is to methodically compare the projected financial costs and benefits of putting the HRMS into place, guaranteeing that the investment is sound from an economic standpoint and offers real value to the company.

Stakeholders can use this analysis to assess whether the system's advantages—such as more efficient operations, reduced paperwork, quicker leave and retirement processing,

increased accuracy, and data-driven HR decisions—outweigh the costs of development, implementation, and maintenance.

COST BENEFIT ANALYSIS						
Development Cost	Quantity	Unit	No. of Months	Cost	Total Amount	
Computers	1	SET	N/A	₱ 24,955.00	₱	24,955.00
Programmer domain	1	N/A	4	₱ 10,000.00	₱	40,000.00
	1	N/A	N/A	₱ 49.00	₱	49.00
Web-hosting	1	N/A	1	₱ 289.00	₱	289.00
				Subtotal	₱	65,293.00
Operational Cost	Quantity	Unit	No. of Months	Cost	Total Amount	
Electricity (Computer)	1	N/A	12	₱ 300.00	₱	3,600.00
Internet Fee	1	N/A	12	₱ 2,499.00	₱	29,988.00
Ink/Toner	1	SET	N/A	₱ 500.00	₱	500.00
Bondpaper	2	REAM	N/A	₱ 250.00	₱	500.00
				Subtotal	₱	34,588.00
				TOTAL COST	₱	99,881.00
Benefit	Quantity	Unit	No. of Months	Cost	Total Amount	
Data Backup	1	N/A	12	₱ 1,700.00	₱	20,400.00
Security of Data	1	N/A	12	₱ 1,250.00	₱	15,000.00
Accessibility	1	N/A	12	₱ 1,300.00	₱	15,600.00
Efficiency of Work Output	1	N/A	12	₱ 3,620.00	₱	43,440.00
				TOTAL BENEFIT	₱	94,440.00

Figure 6. Cost Breakdown

Cost Benefit Analysis						
Details	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
	0	1	2	3	4	5
Present Value Factor	1.00	0.91	0.83	0.75	0.68	0.62
System Cost	65,293.00	34,588.00	38,046.80	41,851.48	46,036.63	50,640.29
PV of System Cost	65,293.00	31,475.08	31,578.84	31,388.61	31,304.91	31,396.98
System Benefit	0.00	94,440.00	103,884.00	114,272.40	125,699.64	138,269.60
PV of System Benefit	0.00	85,940.40	86,223.72	85,704.30	85,475.76	85,727.15
NPV	-65,293.00	54,465.32	54,644.88	54,315.69	54,170.85	54,330.17
Interest Rate	10%					
Total PV of System Benefit	429,071.33					
Total PV of System Cost	222,437.42					
Total NPV	206,633.91					
Last NPV with Negative Cumulative Value						
Payback Period	1		1.1988	YEARS	MONTHS	DAYS
				1	2	12
ROI = (Gain from Investment - Cost of Investment) / Cost of Investment						
Return on Investment	93%					

Figure 7. Cost Benefit Analysis

ERD Model

Figure 8 shows the Entity Relationship Diagram (ERD) of the Bacolod City Local Government Unit (LGU) Human Resource Management System (HRMS). The diagram illustrates the interrelationship between different data elements in the system and how they interact. It is a logical visual representation of the system's structure, enabling the understanding of the flow and interaction of information. The ERD is an essential component in ensuring that all required data is systematically structured, related, and easily accessible, thereby facilitating the creation of a stable and effective system that meets the needs of its users.

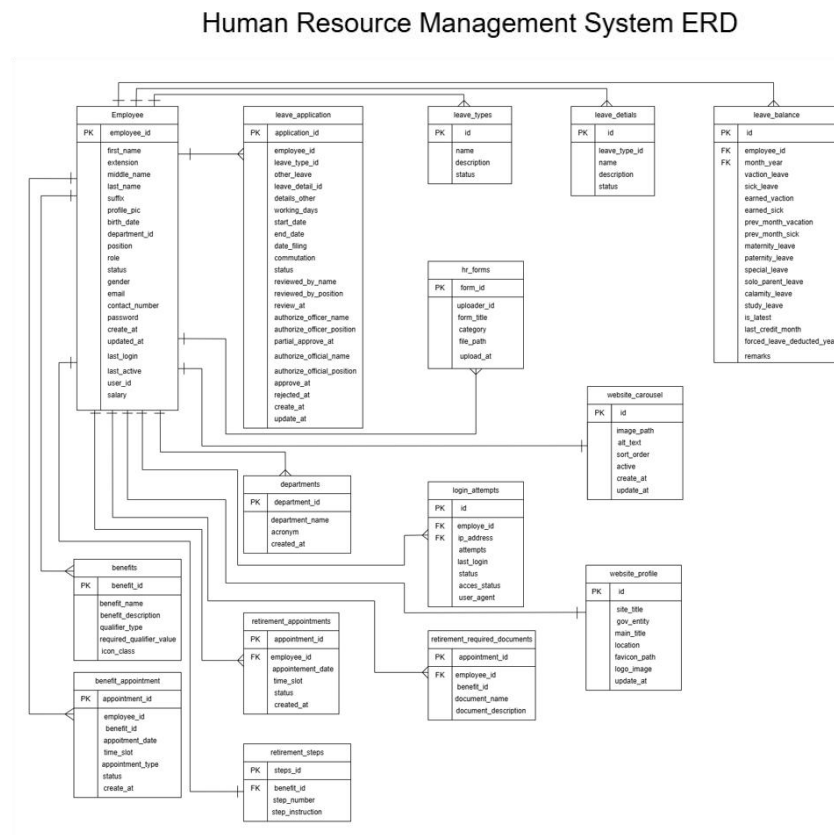


Figure 8. Entity Relationship Diagram

Data Flow Diagram

Figure 10 presents the Context Diagram of the Human Resource Management System (HRMS) of the Bacolod City Local Government Unit, which provides a general overview of the flow of information in and out of the system and indicates the general interactions between the system and its external actors.

At the beginning of the cycle of development, context diagram is crucial in defining a shared understanding of the system participants, constraints, and data exchanges. It compels everyone to consider what the system's boundaries are and where it stands within the organization's context.

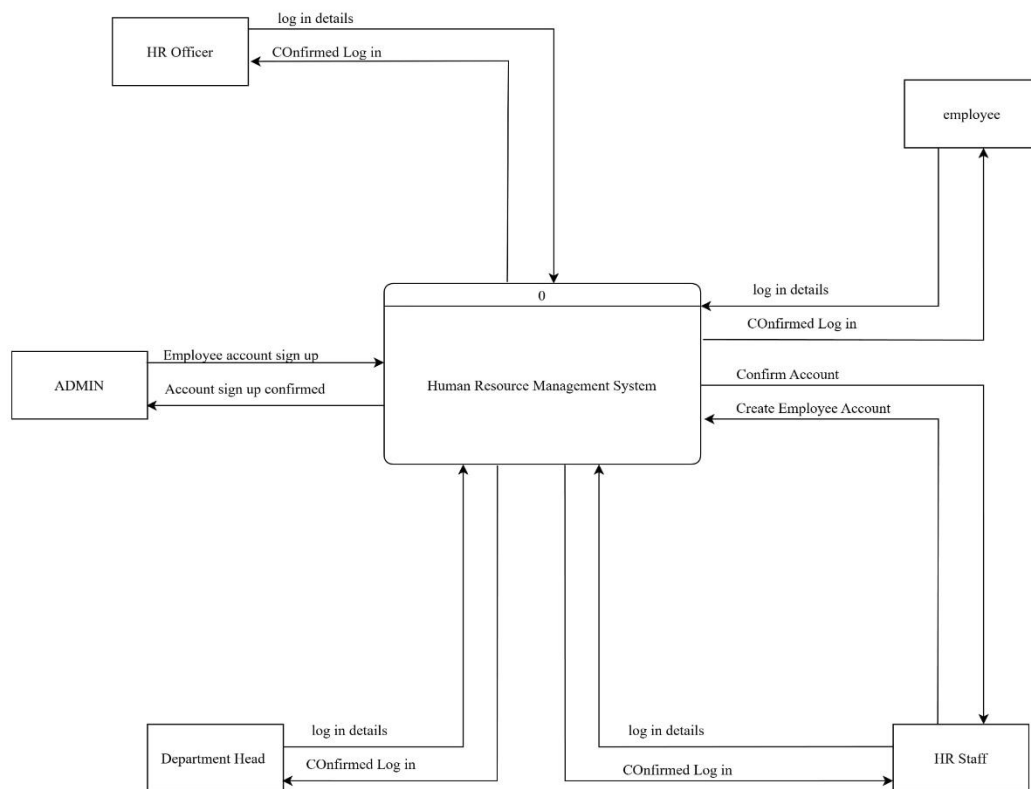


Figure 9. Context Diagram

Figure 10 shows the Data Flow Diagram (DFD) of the Bacolod City LGU's Human Resource Management System (HRMS). This diagram provides a clear picture of how information flows through the system, highlighting key processes, the sources of data, its destinations, storage locations, and the individuals who interact with it. It helps both developers and stakeholders see the bigger picture of how the system operates on a day-to-day basis. By laying out how data flows and connects within the system, the DFD becomes a valuable tool in ensuring everything is organized, efficient, and works as expected.

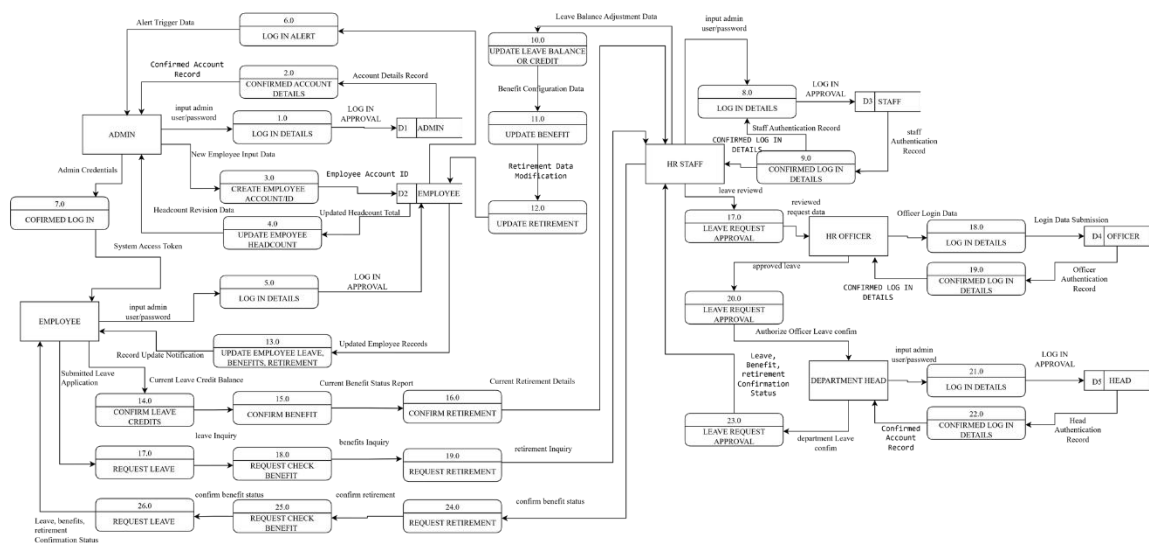


Figure 10. Data Flow Diagram

Data Dictionary

Table 4. employee

Field Name	Description	Type	Length
employee_id	for employee	varchar	11
first_name		varchar	100
extension		varchar	50
middle_name		varchar	100
last_name		varchar	100
suffix		varchar	50
profile_pic		varchar	225
birth_date		date	
department_id		varchar	20
postion		varchar	150
role		enum	
status		enum	
gender		enum	
email		varchar	150
contact_number		varchar	20
password		varchar	255
create_at		timestamp	
update_at		timestamp	
last_login		timestamp	
last_active		timestamp	
user_id		varchar	50
salary		decimal	15,2

Table 5. hr_forms

Field Name	Description	Type	Length
form_id		int	11
upload_id		int	11
form_title		varchar	255
category		enum	50
file_path		varchar	255
uploaded_at		timestamp	

Table 6. leave_application

Field Name	Description	Type	Length
employee_id	for employee	int	11
employee_id		int	11
leave_type_id		int	11
details_other		varchar	255
working_days		int	11
start_date		date	
end_date		date	
date_filing		date	
commutation		enum	
status		enum	
reviewed_by_name		varchar	100
review_by_positon		varchar	100
reviewed_at		datetime	
authorize_officer_name		varchar	100
authorize_officer_position		varchar	100
partial_approved_at		datetime	
authorize_official_name		varchar	100
authorize_official_position		varchar	100
approved_at		datetime	
rejected_at		datetime	
created_at		timestamp	
updated_at		timestamp	

Table 7. leave_balance

Field Name	Description	Type	Length
id		int	11
employee_id		int	11
month_year		date	
vacation_leave		decimal	6,2
sick_leave		decimal	6,2
earned_vacation		decimal	6,2
earned_sick		decimal	6,2
prev_month_vacation		decimal	6,2
prev_month_sick		decimal	6,2
maternity_leave		decimal	6,2
paternity_leave		decimal	6,2
special_leave		decimal	6,2
solo_parent_leave		decimal	6,2
calamity_leave		decimal	6,2
study_leave		decimal	6,2
is_latest		tinyint	1

last_credit_month	date	
forced_leave_deducted_year	int	11
remarks	varchar	255

Table 8. departments

Field Name	Description	Type	Length
department_id		varchar	20
department_name		varchar	255
acronym		varchar	50
created_at		timestamp	

Table 9. benefits

Field Name	Description	Type	Length
benefit_id		Int	11
benefit_name		Varchar	255
benefit_description		text	
qualifier_type		enum	
required_qualifier_value		int	11
icon_class		varchar	50

Table 10. benefits_appointments

Field Name	Description	Type	Length
appointment_id		int	11
employee_id		int	11
benefit_id		int	11
appointment_date		date	
time_slot		time	
appointment_type		varchar	100
status		enum	
created_at		timestamp	

Table 11. leave_details

Field Name	Description	Type	Length
id		int	11
leave_type_id		int	11
name		varchar	255
description		text	
status		tinyintx	1

Table 12. leave_types

Field Name	Description	Type	Length
id		int	11
name		varchar	150
description		text	
status		tinyint	1

Table 13. login_attempts

Field Name	Description	Type	Length
id		int	11
employee_id		varchar	50
ip_address		varchar	100
attempts		int	11
last_login		timestamp	
status		varchar	20
access_status		varchar	20
user_agent		text	

Table 14. retirement_appointments

Field Name	Description	Type	Length
appointment_id		int	11
Employee_id		int	11
appointment_date		date	
time_slot		time	
status		varchar	50
created_at		timestamp	

Table 15. retirement_required_documents

Field Name	Description	Type	Length
document_id		int	11
benefit_id		int	11
document_name		varchar	255
documents_description		text	

Table 15. retirement_steps

Field Name	Description	Type	Length
step_id		int	11
benefit_id		int	11
step_number		int	11
step_instruction		text	

Table 16. website_carousel

Field Name	Description	Type	Length
id		int	11
image_path		varchar	255
alt_text		varchar	255
sort_order		int	11
active		tinyint	1
create_at		datetime	
updated_at		datetime	

Table 17. website_profile

Field Name	Description	Type	Length
id		int	11
site_title		varchar	255
gov_entity		varchar	255
main_title		varchar	255
location		varchar	255
favicon_path		varchar	255
logo_image		varchar	255
update_at		datetime	

Use Case Diagram

A use case diagram represents the different user-system interactions (as seen in Figure 11). A use case diagram illustrates the system's various use cases and user types, and typically, other diagrams accompany it. A use case diagram visually represents the

dynamic character of a system. The researchers use a simplified graphical representation to demonstrate what the system should do in a given use scenario. More diagrams and documentation are needed to give a complete functional and technical perspective of the system.



Figure 11. Use case diagram

CHAPTER IV

PRESENTATION, ANALYSIS, AND DISCUSSION OF RESULTS

This chapter explains and interprets the results of the user survey conducted for the Human Resource Management System and the City Treasures Office.

Presentation

The study outcomes are presented, evaluated, and interpreted in this chapter based on questionnaires distributed by the researchers and completed by 30 participants. The information offered in this chapter focuses on the Human Resource Management System.

The PSSUQ (Post-Study System Usability Questionnaire), a 15-item structured questionnaire, is used by the researcher to deliver the survey. It is used to determine how satisfied users are with the system. The questionnaire is divided into scales. System Usefulness (SYSUSE), Information Quality (INFOQUAL), and Interface Quality (IQ) (INTERQUAL) are three usability scales developed by a group of usability specialists.

Data Analysis

This section presents an analysis of the data collected from respondents in the Human Resource Management System.

Participants of the Study

The respondents for the system were selected from employees of the Human Resource Management Services and the City Treasurer's Office. The total participants was 30 respondents.

Interpretation of Data

Table 18. Scale for Evaluating the Post-Study System Usability Questionnaire (PSSUQ)

Mean Score Range	Interpretation
1.00 – 1.85	Strongly agree
1.86 – 2.71	Agree
2.72 – 3.57	Somewhat agree
3.58 – 4.43	Neither agree or disagree
4.44 -5.29	Somewhat disagree
5.30 – 6.15	Disagree
6.16 – 7.00	Strongly disagree

Results

Table 19. System Usefulness

System Usefulness (SYSUSE)	Mean Score	Standard Deviation	Verbal Interpretation
1. Overall, I am satisfied with how easy it is to use this system.	1.10	0.31	Strongly Agree
2. It was simple to use this system.	1.17	0.38	Strongly Agree
3. I was able to complete the task and scenarios quickly using the system.	1.20	0.41	Strongly Agree
4. I felt comfortable using this system.	1.13	0.35	Strongly Agree
5. It was easy to learn to use this system.	1.17	0.38	Strongly Agree
6. I believe I could become productive quickly using system	1.13	0.35	Strongly Agree
Total Mean	1.15	0.04	Strongly Agree

Table 6 shows an overall mean score of **1.15 (SD = 0.04)** for **System Usefulness**, interpreted as **Strongly Agree**. According to Atulya Nagar et al. (2024), the high Degree of agreement that the website is easy to use and comprehend directly translates into user confidence that the system is effective and efficient for the task.

Table 20. Information Quality

Information Quality (INFOQUAL)	Mean Score	Standard Deviation	Verbal Interpretation
1. The system gave error messages that clearly told me how to fix problems.	1.57	1.10	Strongly Agree
2. Whenever I made a mistake using the system, I could recover easily and quickly.	1.33	0.70	Strongly Agree
3. The information provided with this system (online help, documentation) was clear.	1.37	0.38	Strongly Agree
4. It was easy to find the information I needed.	1.17	0.38	Strongly Agree
5. The information was effective in helping me complete the task and scenarios	1.27	0.52	Strongly Agree
6. The organization of information on the system screens was clear.	1.20	0.48	Agree
Total Mean	1.31	0.39	Strongly Agree

Table 7 shows the mean result of Information Quality as 1.31, interpreted as Strongly Agree. According to Azwan Abd Aziz (2024), the findings reveal that prioritizing high information quality is non-negotiable, as it is a critical lever for improving operational efficiency and driving stronger user satisfaction scores.

Table 21. Interface Quality

Human Resource Management System

Interface Quality (INTERQUAL)	Mean Score	Standard Deviation	Verbal Interpretation
1. This interface of this system was pleasant.	1.20	0.41	Strongly Agree
2. I like using the interface of this system.	1.17	0.38	Strongly Agree
3. This system has all the functions and capabilities I expect it to have	1.37	0.67	Strongly Agree
Total Mean	1.24	0.46	Strongly Agree

Table 12 shows the mean result of Interface Quality as 1.24, interpreted as Strongly Agree. According to William P. Rey et al. (2024), the primary objective is to minimize cognitive load and maximize task completion rates by providing users with a user-friendly and efficient interface.

Table 22. Mean Results of Post-Study System Usability Questionnaire (PSSUQ)

Criteria	N	Mean	Interpretation
System Usefulness (SYSUSE)	20	1.15	Strongly Agree
Information Quality	20	1.31	Strongly Agree
Interface Quality	20	1.24	Strongly Agree
Overall	20	1.23	Strongly Agree

The average System Usefulness (SYSUSE) value is 1.15, interpreted as Strongly Agree. According to Atulya Nagar et al. (2024), the high Degree of agreement that the website is easy to use and comprehend directly translates into user confidence that the system is effective and efficient for the task.

The average Information Quality (INFOQUAL) value is 1.31, interpreted as Strongly Agree. According to Azwan Abd Aziz (2024), the findings reveal that

prioritizing high information quality is non-negotiable, as it is a critical lever for improving operational efficiency and driving stronger user satisfaction scores.

The average Interface Quality (INTERQUAL) value is 1.24, interpreted as Strongly Agree. According William P. Rey et al. (2024), Interface Quality refers to user-friendliness that provides a comprehensive range of capabilities for specifying problems and processing, analyzing, and presenting findings. There should be no extraneous or distracting information on the interface. Displays should be as consistent as feasible. Users should not be required to recall information when they go from one portion of the interface to another. The instructions should not be required to be memorized by the users. The system's operating instructions should be visible.

Overall, the mean result is 1.23, interpreted as Strongly Agree. According to William P. Rey (2024), the study evaluates the system's usability using the System Usability Scale (SUS) and the Post-Study System Usability Questionnaire (PSSUQ) (OVERALL PSSUQ).

This result means that the Human Resource Management System can help the different department offices in Bacolod City by means of providing the basic HR service information such as real-time leave balance monitoring for the employee, monitoring the incoming retirees, providing complete data on leave entitlements and retirement eligibility, generating complete reports needed by the HR staff, monitoring the status of

all leave applications, and tracking the employment records of every individual, especially the employees.

CHAPTER V

SUMMARY OF FINDINGS, CONCLUSION AND RECOMMENDATION

Summary of Findings

The researchers aimed to develop a Human Resource Management System (HRMS) that can provide an effective and efficient record management system for quick retrieval and faster decision-making regarding employee entitlements.

The results of the system evaluation indicate that the metrics for System Usefulness (SYSUSE), Interface Quality, and Overall Results are all interpreted as "Strongly Agree." As a result, the system's design and actions are more focused on addressing people's demands in their employment, particularly in managing leave, benefits, and retirement processes. The findings also successfully address how technology might be integrated into users' social and physical settings and how the system is used efficiently. Crucially, the system requires less user input to produce the desired result, thereby increasing overall efficiency.

Conclusion

Based on the study's consistently positive findings, users were unanimous in concluding that the developed Human Resource Management System (HRMS) will be

highly beneficial to the department offices within the Bacolod City LGU if implemented. By automating and centralizing routine administrative tasks, the system effectively supplies the required information instantly, enabling both employees and HR personnel to concentrate more on their core job functions rather than manual processing. Furthermore, users strongly agreed that employing the HRMS will provide substantial organizational benefits through the software's capabilities, including high ease of use, overall cost-effectiveness, significant time and resource savings, improved data accuracy and security, and enhanced employee relationships through transparent and efficient service delivery.

Recommendations

1. For further improvement, the system can integrate a facial recognition attendance the system to improve the deduction of late and absents to the leave balance.
2. The system may include a mobile application version to allow employees to view attendance, file leaves, and receive HR notifications conveniently.
3. To enhance the HRMS, integrating payroll computation directly with attendance and leave records is recommended for faster and more accurate payroll processing.
4. *To enhance system monitoring, an audit trail feature is recommended to record all critical actions performed within the HRMS for transparency and accountability.*

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APPENDICES

Transmittal Letter

Minutes of the Capstone

Consolidated Results of Testing

Adviser Acceptance Form

Adviser Endorsement Form

Compliance Matrix/Sheet

Requirements for Oral Defense

Grammarians Certificate (Grammarly)



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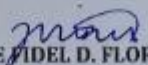
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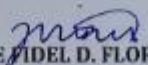
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